

Mohit Agrawal

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Location: Pune, India (Willing to relocate)

<https://mohit-ai-portfolio.netlify.app/>

PROFESSIONAL SUMMARY

AI Engineer / Data Scientist with 3.5 years of experience in data analytics, **machine learning, Generative AI, NLP, and AML/Fraud analytics**. Experienced in developing **CDD and EDD solutions, entity resolution, adverse media screening**, risk intelligence, and automated financial crime investigation workflows using **Python, LLMs, AWS Bedrock, FastAPI, and MCP-based integrations**. Strong hands-on experience in predictive modeling, data analysis, **LLM-powered applications, RAG, and Agentic AI**, with a focus on building scalable, **production-ready AI solutions**.

EDUCATION AND HONORS

International University of Applied Sciences, Berlin, Germany

Master Of Science (M.S.) in Computer Science - Pursuing

Sant Gadge Baba Amravati University, Amravati, Maharashtra

Bachelor of Engineering (B.E.) in Computer Science and Engineering

[July 2018-July 2022]

TECHNICAL SKILLS

Programming Languages : Python, SQL, Java, C++

Machine Learning : Regression, Classification, Clustering, Time Series Forecasting, Feature Engineering, Model Evaluation, Hyperparameter Tuning

Deep Learning : TensorFlow, PyTorch, Neural Networks, LSTM, Transformers

Natural Language Processing & Generative AI : OpenAI GPT, LangChain, Hugging Face, BERT, spaCy, Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, FAISS, Vector Databases, Agentic AI, Tool Calling

Frameworks & Deployment : FastAPI, Flask, Streamlit, Docker, Git, REST APIs

Data Analytics & Visualization : Power BI, Tableau, Excel

Databases : PostgreSQL, MySQL

Agentic AI: ReAct, Tool Calling, MCP (Model Context Protocol)

PROFESSIONAL EXPERIENCE

Data Scientist – AML / Fraud Analytics

Solytics Partners | Pune, India |

August 2026 – Present

- Develop **AI/ML and Generative AI-driven CDD and EDD solutions** to automate **customer due diligence, enhanced due diligence**, and financial crime investigation workflows.
- Build entity resolution and **screening workflows** for individuals and organizations by **analyzing names, jurisdictions, registration identifiers, and other entity attributes to accurately identify and validate entities**.
- Integrate **adverse media, OpenCorporates, and Unified Search Service (USS)**

data sources to research entities, **identify potential risk indicators**, and support AML/Fraud investigations.

- Develop **automated CDD/EDD investigation and report-generation pipelines** using Python, FastAPI, LLMs, **AWS Bedrock**, and **MCP-based** tool integrations, consolidating **research findings and risk signals** into structured reports.
 - Apply data analytics, machine learning, and LLM-based techniques to identify **suspicious patterns**, **assess entity/transaction risk**, and improve the efficiency and consistency of **AML and Financial Crime Compliance (FCC) processes**.
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Software Engineer (AI Developer)

Highbar Technocrat (Client : Maharashtra Metro Rail Corporation Ltd.) – Nagpur

March 2025 – August 2026

- Developed and maintained 10+ operational dashboards for **AFC, OCC, CRM, revenue, ridership, and station performance using Power BI**.
 - Automated daily **data ingestion, validation, and reporting workflows** using SQL and Python.
 - Collaborated with cross-functional teams to deliver **forecasting models for passenger demand and revenue planning**.
 - Built **anomaly detection logic** to identify unusual **passenger flow** and operational deviations.
 - Designed **KPI dashboards** for senior management supporting operational and **business decisions**
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Database Operations Analyst

Bamboobox, Bangalore (A product-based company)

January 2025 – March 2025

- Conducted **data enrichment and data mining** using tools like **LSN, Apollo, and Lusha**.
 - Cleaned, processed, and validated **large datasets** to improve **lead generation and client prospecting**.
 - Optimized **data pipelines** to improve data quality and usability.
 - Ensured **data integrity, quality and compliance and governance standards**.
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Data Analyst & Machine Learning Engineer

Samatrix Consulting Pvt. Ltd., Gurgaon

July 2024 – December 2024

- Built **time-series and regression-based forecasting models** to predict business trends.
 - Cleaned and processed large multi-source datasets using **Python (Pandas, NumPy)**.
 - Applied **feature engineering techniques** to improve forecast reliability and stability.
 - Built **Power BI and Tableau dashboards** to communicate forecasting insights.
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Junior Data Analyst

Starlink Technology Pvt. Ltd., Pune

March 2023 – July 2024

- Built **predictive and demand forecasting models** using Python, SQL, and scikit-learn.
- Conducted **EDA, feature engineering, and hypothesis testing** to improve

- forecast accuracy.
- Developed **trend and time-series models** using large historical datasets.
- Created automated **Tableau dashboards** to track trends and forecast outputs.
- Supported **data-driven planning** through reliable forecasting insights..

PROJECTS

AI Resume Intelligence System (ATS + Semantic Matching)

Tech Stack: Python, FastAPI, Sentence Transformers (BERT), spaCy, scikit-learn, PDFMiner, Docker, Hugging Face

- Built an **AI-powered Resume Intelligence System** to parse **PDF resumes**, calculate **ATS scores**, and perform **semantic job matching** using **NLP and transformer embeddings**.
- Developed **resume parsing modules** to extract candidate information **including contact details, skills, education, and experience** from unstructured PDF resumes.
- Implemented **BERT-based semantic similarity** using **Sentence Transformers** to compare resumes with **job descriptions and generate match scores**.
- Designed an **ATS scoring engine** to evaluate resume-job fit **and identify matched and missing skills**.
- Exposed functionality through **FastAPI REST APIs** and generated structured **JSON responses** for **downstream recruitment** workflows.
- Containerized the application **using Docker** and **deployed it on Hugging Face Spaces** for consistent and scalable execution.

Retrieval-Augmented Generation (RAG) based PDF Question Answering System

Tech Stack: Python, FastAPI, Streamlit, LangChain, Hugging Face, FAISS, Docker

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** application to **answer natural language questions** from PDF documents.
- Developed a document ingestion pipeline using **LangChain, PyPDFLoader, and Recursive Character Text Splitter** for semantic document processing.
- Generated **semantic embeddings using Hugging Face Sentence Transformers** and **indexed documents in FAISS** for efficient vector similarity search.
- Implemented **context-aware retrieval** to fetch the most relevant document chunks before **generating grounded AI responses**.
- Designed prompt templates to improve **response relevance** and **reduce hallucinations**.
- Developed **REST APIs using FastAPI** and an **interactive Streamlit frontend** for document upload and **question answering**.
- **Containerized the application using Docker** and structured the project for modular, production-style development.

AI Inventory Management Agent

Tech Stack: Python, FastAPI, React, LangChain, Groq (Llama 3.3 70B), PostgreSQL, SQLAlchemy, Docker

- Built an end-to-end **AI Inventory Management Agent** capable of interpreting **natural language requests** and executing **multi-step inventory workflows**.
- Implemented ReAct-style reasoning with tool calling for **inventory search, low-stock detection, order history retrieval, order creation, status updates**, and order cancellation.
- Integrated **PostgreSQL with SQLAlchemy** for inventory and order management and exposed **REST APIs using FastAPI**.

- **Implemented input validation** and structured exception handling **to improve API reliability** and user experience.
- Containerized the **frontend and backend using Docker** and Docker Compose for consistent local deployment.

CERTIFICATIONS

- **Google Data Analytics** Specialization (Google)
- **Data Analytics and Visualization** Job Simulation (Accenture North America)
- **Data Analytics Consulting** Virtual Internship (KPMG)
- **Artificial Intelligence** Virtual Experience Internship (Cognizant)

ACHIEVEMENTS

- Built multiple AI and ML applications using **LangChain, FastAPI and OpenAI APIs**.
- **Developed production dashboards** used for operational monitoring.
- Active **Github contributor** with projects in **Gen AI and Machine Learning**
- Developed **AI powered automation and analytics solutions** for business reporting and decision support

PROFILE LINK

[LinkedIn](https://www.linkedin.com/in/mohit-agrawal45) :- <https://www.linkedin.com/in/mohit-agrawal45>

[GitHub](https://github.com/Mohhitt45) :- <https://github.com/Mohhitt45>
